

CS215 Assignment 4

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Contents

1	Parking Lot Problem	3
1.1	Part A	3
1.2	Part B	3
1.3	Part C	5
2	Forecasting on a Real World Dataset	6
2.1	Task 1	6
2.1.1	Part (a)	6
2.1.2	Part (b)	6
2.1.3	Part (c)	7
2.2	Task 2	7
2.3	Task 3	8

1 Parking Lot Problem

1.1 Part A

Check code in Q1a.ipynb

Answers:

The values of **MASE**(Mean absolute scaled error) and **MAPE**(Mean absolute percentage error) for the test data(split from given data) are:

1. $\text{MASE} = 0.7600$
2. $\text{MAPE} = 0.0509$

The predicted values of the total number of vehicles entering the parking per day, for next 7 days is as follows:

1. 14th November: 840.392345
2. 15th November: 862.900817
3. 16th November: 813.441560
4. 17th November: 923.822111
5. 18th November: 840.294375
6. 19th November: 813.022432
7. 20th November: 857.036687

1.2 Part B

Check code in Q1b.ipynb

Answers:

The values of **MASE**(Mean absolute scaled error) and **MAPE**(Mean absolute percentage error) for the test data(split from given data) are:

1. $\text{MASE} = 0.0921$
2. $\text{MAPE} = 0.0062$

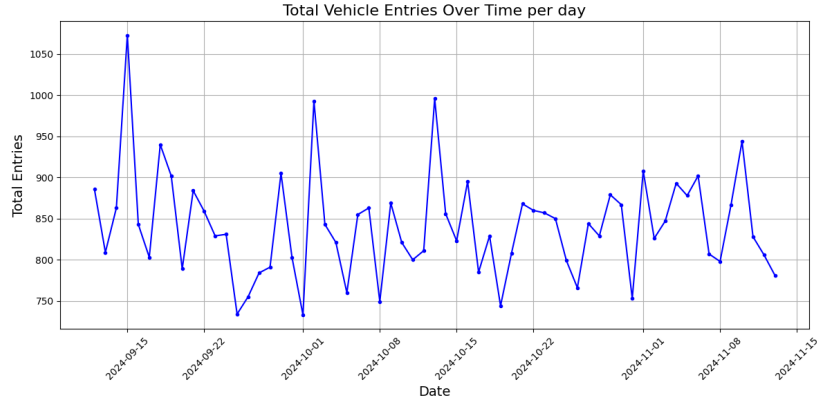


Figure 1: Total vehicle entries per day(Time series in Part A)

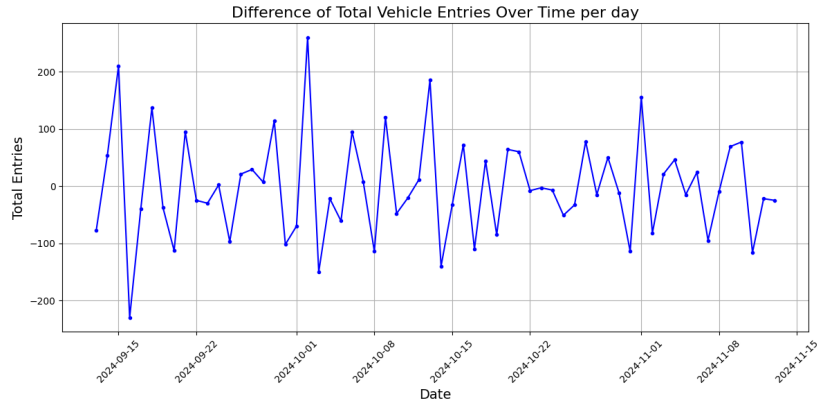


Figure 2: Difference Series($d = 1$) in Part A

The predicted values of the average times in minutes spent in the mall by a vehicle entering on a particular day for the next 7 days is as follows:

1. 14th November: 339.567819
2. 15th November: 360.657282
3. 16th November: 368.624311
4. 17th November: 362.162941
5. 18th November: 343.220077

6. 19th November: 319.990320

7. 20th November: 301.078917

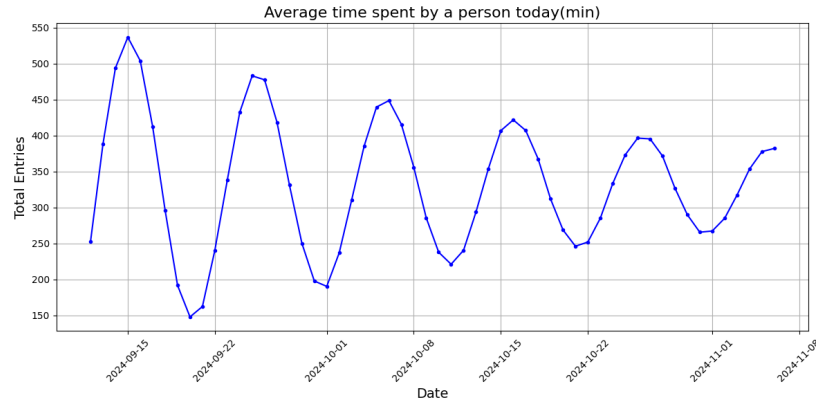


Figure 3: Time Series in Part C(very smooth and has no trend)

1.3 Part C

Check code in Q1c.ipynb

Answers:

In this report, I have implemented two outlier smoothing routines to preprocess the data before feeding it into the model. These routines are:

1. **Z-Score Outlier Smoothing:** This method calculates the Z-scores of the data points, which indicates how many standard deviations each point is from the mean. Any data point with an absolute Z-score greater than a specified threshold (commonly set to 3) is considered an outlier. These outliers are then replaced with the mean of the dataset to mitigate their influence on the analysis.
2. **IQR Outlier Smoothing:** This method utilizes the interquartile range (IQR) to identify outliers. The IQR is computed as the difference between the third quartile (Q3) and the first quartile (Q1). Data points that lie below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ are classified as outliers. Similar to the Z-score method, these outliers are replaced with the mean of the dataset.

The following table summarizes the Mean Absolute Percentage Error (MAPE) and Mean Absolute Scaled Error (MASE) for each part of the analysis, using the Z-score and IQR outlier smoothing methods.

Table 1: Performance Metrics for Outlier Smoothing Methods

Part	Method	MAPE	MASE
Part 1	Z-Score	0.0500	0.7442
Part 1	IQR	0.0533	0.7911
Part 2	Z-Score	0.0062	0.0921
Part 2	IQR	0.0062	0.0921

2 Forecasting on a Real World Dataset

2.1 Task 1

2.1.1 Part (a)

Code in Q2a.ipynb

Hyperparameters are chosen :

- for p and q , the first significant spike drop observed in the **ACF** and **PACF** plots respectively is considered as optimal values
- As for d , **adfuller** function is used to determine optimal d value

2.1.2 Part (b)

Code in Q2b.ipynb

Best working prompt strategy :

I have a time series dataset with 127 data points that represent monthly operational metrics for a major Indian airline from 2013. It includes information on the number of passengers traveled. The data follows an unknown model of time series. Based on this data, predict the next 12 values in the sequence and use the best model for the time series.

Please provide a list of 12 predicted values that could follow this series.
Here is the data: [String of data generated using a python script]

Evaluation of the prompt :

Data Frame : 7958524.0, 8050154.0, 8141784.0, 8233414.0, 8325044.0, 8416674.0, 8508304.0, 8599934.0, 8691564.0, 8783194.0, 8874824.0, 8966454.0
MAPE of the dataframe predicted : 5.92

2.1.3 Part (c)

Code in Q2c.ipynb

Global Model used : Neural Prophet

Dataframe : 7419019.5, 7914882.0, 8173941.0, 8574955.0, 8255045.5, 8246122.5, 8498941.0, 8273163.0, 8570758.0, 8351914.0, 8536465.0, 8747979.0
MAPE of predicted data frame : 4.00

2.2 Task 2

Solution:

Why MAPE may not be a good metric to evaluate the forecasts:

MAPE (Mean Absolute Percentage Error) measures forecast accuracy as a percentage error. While MAPE can be a useful metrics, it has notable limitations. Here are the primary reasons:

i. Sensitivity to Small Actual Values:

- MAPE calculates the percentage error by dividing the absolute error by the actual value for each data point.
- If there are months with low passenger numbers (e.g., off-peak periods or months affected by events like COVID etc..) which results in low actual values, thus turns even small forecast errors to result in large percentage errors, which can distort the metric and make the forecast appear worse than it is.

ii. Underrepresentation of Peak Demand:

- In fleet and workforce planning, airlines need to ensure they can handle peak demand periods, not just average demand.
- MAPE does not account for peak error directly and treats all errors equally, potentially underestimating the importance of errors during high-demand periods. An accurate understanding of peak periods is crucial since insufficient resources during these times can lead to significant operational disruptions and customer dissatisfaction.

iii. Focus on Total Demand (Fleet Planning):

- Fleet requirements are typically dictated by total expected passenger numbers rather than percentage deviations. If the goal is to meet cumulative demand accurately, MAPE may not align well with this objective, as it emphasizes percentage accuracy rather than absolute volume.

Suggested Alternative Metrics:

i. Weighted Absolute Percentage Error (WAPE):

- WAPE is the sum of absolute errors divided by the sum of actual values, avoiding the problem of exaggerated errors at low values. This can give a more accurate picture of overall forecast performance without inflating errors in low-demand periods.

ii. Root Mean Squared Error (RMSE):

- RMSE is effective for fleet planning because it emphasizes larger errors more heavily, providing a clearer picture of significant forecast deviations. By focusing on total volume error rather than percentage error, it better aligns with fleet planning needs, where the overall passenger count is essential.

2.3 Task 3

Solution:

To compare whether there's a structural break in the mean of this series before and after the COVID-19 period, we need a statistical test that can determine if there's a significant change in mean from the pre-COVID period to the post-COVID period.

Testing Method:

- Since we're looking at passenger counts specifically, we can perform a structural break test to see if there's a significant difference in the mean of ΔY (first difference in monthly "Passengers Carried") across these periods.

Suitable Test: Chow Test for Structural Break:

- **Purpose:** The Chow Test is a common test to detect structural breaks at a specific point in time in time-series data.
- **How it works:**
 - The Chow test splits the time series at a given point—in this case, just before December 2019, with the post-COVID period starting after January 2022.
 - The test compares the mean and other statistical properties of each segment, calculating an F-statistic to see if the model's parameters (mean, in this case) are significantly different across the two time periods.
- **Assumptions:**
 - The Chow Test assumes normally distributed errors, which fits well here as the first-differenced series is assumed stationary with known variance σ^2
- **Interpretation:**
 - If the test statistic is significant, it suggests a structural break, indicating that the mean of ΔY likely shifted post-COVID.
 - If the t-statistic is not significant, it suggests that the mean of monthly changes in passenger counts did not differ substantially, implying that the demand pattern may not have changed in terms of average month-to-month fluctuations.

Alternative: Two-Sample t-Test for Difference in Means:

- **Purpose:** If the primary objective is to test for a change in the mean of ΔY across pre- and post-COVID periods, a Two-Sample t-Test for the difference in means can serve as a simpler alternative to the Chow Test.

- **How it works:**

- Calculate the mean of ΔY for both periods.
- Use the t-test to compare the means. A significant t-statistic would indicate a difference in the average monthly change in passenger numbers, suggesting an impact from COVID-19

- **Interpretation:**

- A significant t-statistic indicates that the average monthly change in passengers carried differs between the pre-COVID and post-COVID periods, supporting the hypothesis that COVID-19 structurally affected passenger demand.

Final thoughts: the Chow test would be particularly suitable because it directly tests for a change in the relationship (mean in this case) before and after a specific point (pre-COVID and post-COVID periods)